EXP -1 : ANALYSIS OF BASIC SIGNAL OPERATORS

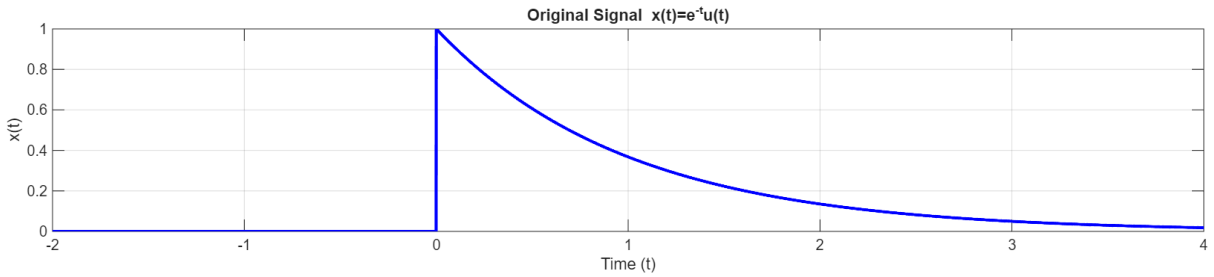
TASK 1

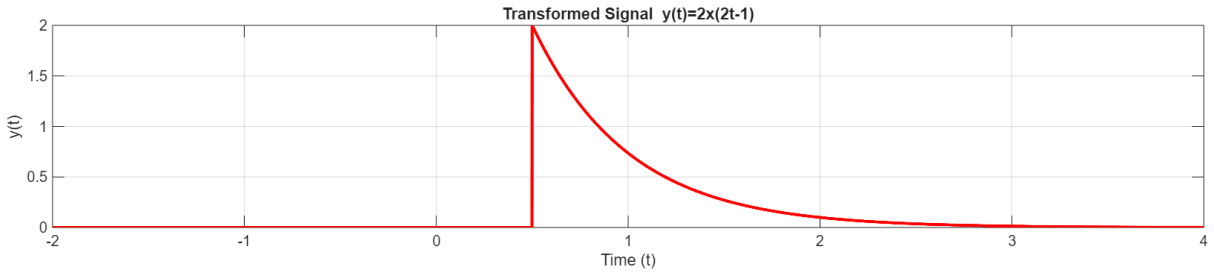
INPUT CODING

```
clc;
clear;
close all;
% Time vector
t = -2:0.001:4;
% Unit step function
u = double(t>=0);
% Original signal x(t)=exp(-t)u(t)
x = exp(-t).*u;
% Transformed signal y(t)=2x(2t-1)
t1 = 2*t - 1;
u1 = double(t1>=0);
y = 2*exp(-(t1)).*u1;
% Plotting
figure;
subplot(2,1,1)
plot(t,x,'b','LineWidth',2)
grid on
xlabel('Time (t)')
ylabel('x(t)')
title('Original Signal  x(t)=e^{-t}u(t)')

subplot(2,1,2)
plot(t,y,'r','LineWidth',2)
grid on
xlabel('Time (t)')
ylabel('y(t)')
title('Transformed Signal  y(t)=2x(2t-1)')
```

OUTPUT





Expected Observation

- Time Compression by factor **2**
- Delay of **0.5 units**
- Amplitude doubled

TASK 2

Input coding

```
clc;
clear;
close all;
% Time vector
t = -3:0.001:3;
% Original pulse
x = double(t>=-1) - double(t>=1);
% Shifted signal x(t-1)
x_shift = double((t-1)>=-1) - double((t-1)>=1);
% Reversed signal x(-t)
x_reverse = double((-t)>=-1) - double((-t)>=1);
% Compressed signal x(2t)
x_compress = double((2*t)>=-1) - double((2*t)>=1);
% Plotting
figure;

subplot(2,2,1)
plot(t,x,'LineWidth',2)
grid on
title('x(t)')
xlabel('Time')
ylabel('Amplitude')

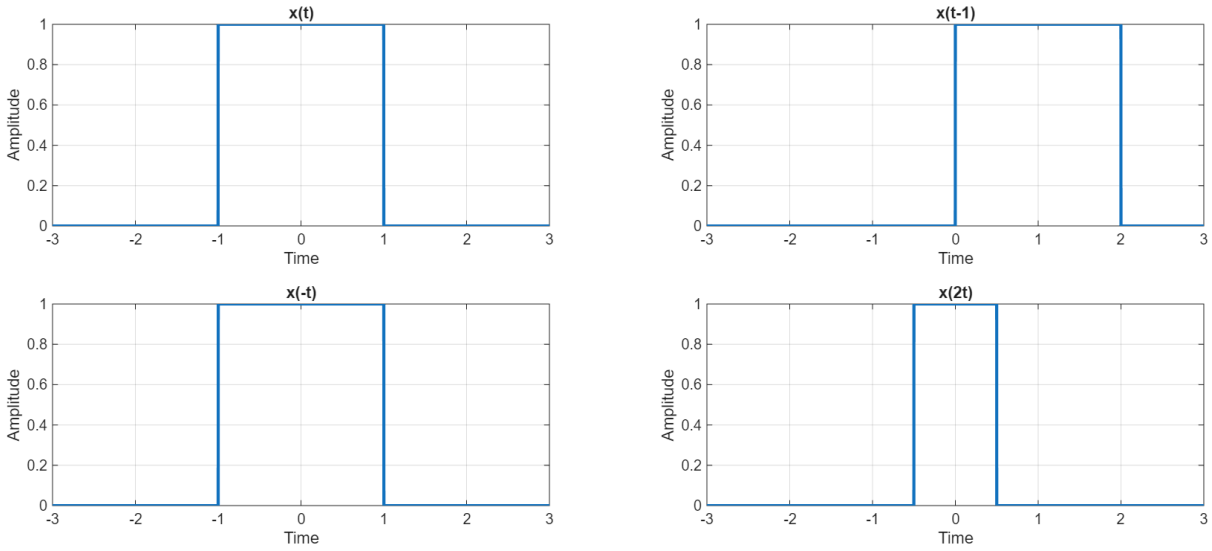
subplot(2,2,2)
plot(t,x_shift,'LineWidth',2)
grid on
title('x(t-1)')
xlabel('Time')
ylabel('Amplitude')

subplot(2,2,3)
plot(t,x_reverse,'LineWidth',2)
grid on
```

```
title('x(-t)')
xlabel('Time')
ylabel('Amplitude')

subplot(2,2,4)
plot(t,x_compress,'LineWidth',2)
grid on
title('x(2t)')
xlabel('Time')
ylabel('Amplitude')
```

OUTPUT



Signal	**Effect**
x(t)	Original pulse
x(t−1)	Shifted right by 1
x(−t)	Time reversal
x(2t)	Pulse width reduced by half (compression)

TASK 3(CONTINUOUS SIGNAL)

```
clc;
clear;
close all;
% Time vector
t = -2:0.001:2;
% Original signal
```

```
x = sin(2*pi*t);



% Amplitude scaling
x_amp = 2*x;
% Time shift
x_shift = sin(2*pi*(t-0.5));
% Time reversal
x_reverse = sin(2*pi*(-t));

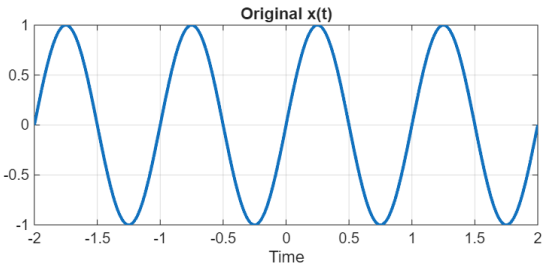
% Plotting
figure;

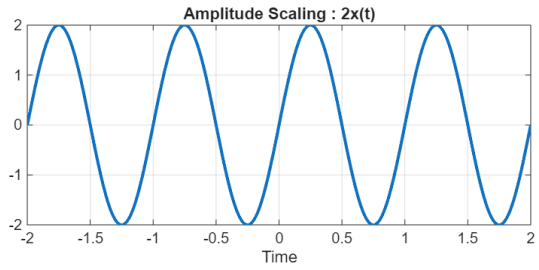
subplot(2,2,1)
plot(t,x,'LineWidth',2)
grid on
title('Original x(t)')
xlabel('Time')

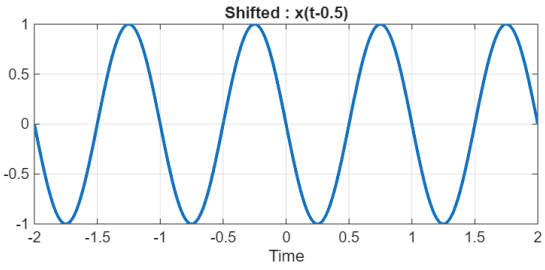
subplot(2,2,2)
plot(t,x_amp,'LineWidth',2)
grid on
title('Amplitude Scaling : 2x(t)')
xlabel('Time')

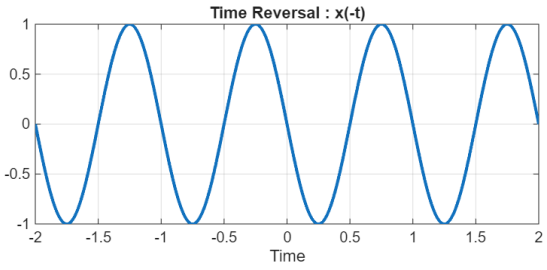
subplot(2,2,3)
plot(t,x_shift,'LineWidth',2)
grid on
title('Shifted : x(t-0.5)')
xlabel('Time')

subplot(2,2,4)
plot(t,x_reverse,'LineWidth',2)
grid on
title('Time Reversal : x(-t)')
xlabel('Time')
```









Operation	**Observation**
2x(t)	Amplitude doubled
x(t−0.5)	Right shift by 0.5 units
x(−t)	Time reversal

TASK 3 (DISCRETE SIGNAL)

```
clc;
clear;
close all;

% Index vector
n = -4:4;

% Original sequence
x = (-1).^n;

% Amplitude scaling
x_amp = 2*x;

% Right shift by 2
x_shift = (-1).^(n-2);

% Time reversal
x_reverse = (-1).^(-n);

% Plotting

figure;
```

```
subplot(2,2,1)
stem(n,x,'filled','LineWidth',1.5)
grid on
title('Original x[n]')
xlabel('n')

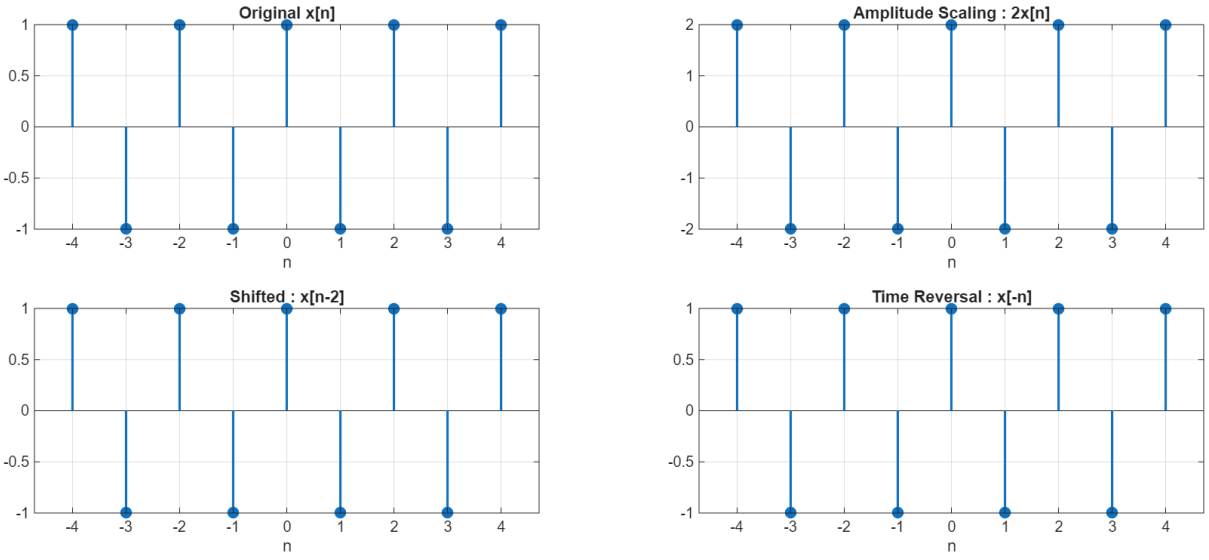
subplot(2,2,2)
stem(n,x_amp,'filled','LineWidth',1.5)
grid on
title('Amplitude Scaling : 2x[n]')
xlabel('n')

subplot(2,2,3)
stem(n,x_shift,'filled','LineWidth',1.5)
grid on
title('Shifted : x[n-2]')
xlabel('n')

subplot(2,2,4)
stem(n,x_reverse,'filled','LineWidth',1.5)
grid on
title('Time Reversal : x[-n]')

xlabel('n')
```

OUTPUT



Operation	**Observation**
2x[n]	Amplitude doubled
x[n−2]	Right shift by 2 samples

Operation	Observation
x[−n]	Sequence reversed (for this specific signal, it appears identical because ((-1)^{-n} = (-1)^n))